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ASSIGNMENT-1

Subject: Computer Networks

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Submitted by:

Name: Bharti Mehra

UID: 24BAI70051

Section: 24AIT_KRG-1_B

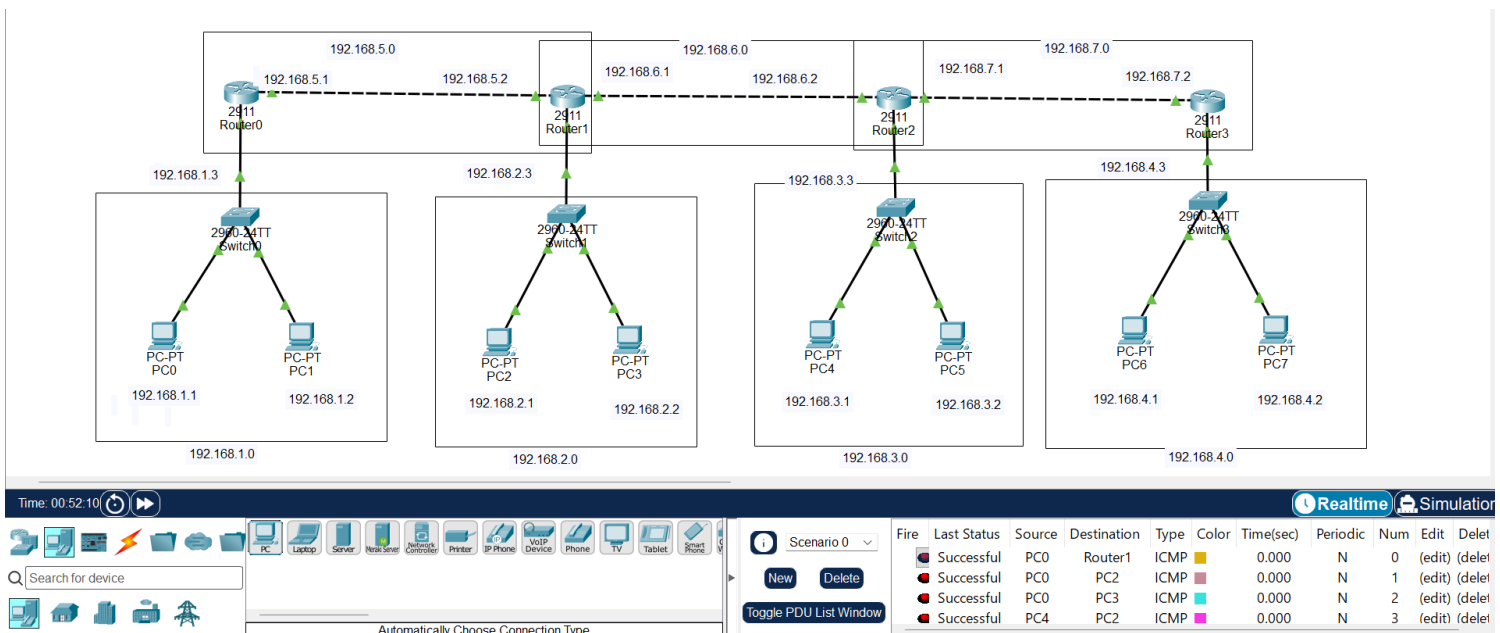
Submitted to:

Faculty Name: Alok Sir

Assignment – 1

Q1. Develop Static Routing with 4 PC's, 4 Switches & 8 PC's. Make sure to highlight the Networks with their network ID's. IP Addresses should be Written in text format as well. (6 M)

Ans. A fully configured 4-router static network was developed through the Cisco Packet Tracer. Following is a screenshot of the network. All subnets are highlighted with their Network IDs and individual IP addresses, with end-to-end connectivity successfully tested across all devices.



Q2. What is OSI model? What are the different layers of OSI model? Write the functionality, Data format and Protocols used at each layer. (3 M)

Ans. OSI (Open Systems Interconnection) Model is a 7-layer conceptual framework developed by the International Organization for Standardization (ISO) to explain how data is transmitted from one computer to another over a network.

Each layer performs a specific function and communicates with the layer directly above and below it. The seven layers are:

1. Application Layer: This is the 7th and highest layer of the OSI model.

- a. Functionality:
 - i. Provides network services directly to user applications.
 - ii. Allows applications to communicate over a network.
 - iii. Provides services such as web browsing, email and file transfer
 - b. Data Format: Data/Message
 - c. Protocols:
 - i. HTTP/HTTPS – Web browsing
 - ii. FTP – File transfer
 - iii. SMTP – Sending emails
 - iv. DNS – Domain name resolution
 - v. DHCP – Assigning IP addresses
 - vi. SSH – Secure remote access
2. Presentation Layer: This is the 6th layer. It acts as a translator between the application and the network.
- a. Functionality:
 - i. Data translation – Converts data into a format that different systems can understand.
 - ii. Encryption and decryption – Protects data from unauthorized access.
 - iii. Compression and decompression – Reduces the size of data.
 - iv. Handles different data representations and character encoding.
 - b. Data Format: Data
 - c. Protocols:
 - i. SSL/TLS – Encryption
 - ii. JPEG – Image format
 - iii. PNG – Image format
 - iv. MPEG – Multimedia format
 - v. ASCII
 - vi. Unicode
3. Session Layer: This is the 5th layer. It manages communication sessions between two applications.
- a. Functionality:
 - i. Establishes a session between two devices/applications.
 - ii. Maintains the session.

- iii. Synchronizes communication.
 - iv. Terminates the session.
 - v. Provides checkpoints for recovering interrupted communication.
 - b. Data Format: Data
 - c. Protocols:
 - i. NetBIOS
 - ii. RPC (Remote Procedure Call)
 - iii. SIP (Session Initiation Protocol)
 - iv. PPTP
- 4. Transport Layer: This is the 4th layer. It provides end-to-end communication between applications on different devices.
 - a. Functionality:
 - i. Breaks large data into smaller pieces.
 - ii. Provides segmentation and reassembly.
 - iii. Provides error control.
 - iv. Provides flow control.
 - v. Provides reliable or unreliable delivery.
 - vi. Uses port numbers to identify applications.
 - vii. Provides end-to-end communication.
 - b. Data Format: Segment (for TCP), Datagram (For UDP)
 - c. Protocols:
 - i. TCP – Transmission Control Protocol
 - 1. Connection-oriented
 - 2. Reliable
 - 3. Guarantees ordered delivery
 - 4. Uses acknowledgements and retransmission
 - ii. UDP – User Datagram Protocol
 - 1. Connectionless
 - 2. Faster than TCP
 - 3. Does not guarantee delivery
 - 4. Used where speed is more important than reliability
- 5. Network Layer: This is the 3rd layer. It is responsible for delivering data from the source network to the destination network.

- a. Functionality:
 - i. Provides logical addressing using IP addresses.
 - ii. Performs routing.
 - iii. Determines the best path for data.
 - iv. Forwards packets between different networks.
 - v. Handles packet delivery across multiple networks.
 - b. Data Format: Packet
 - c. Protocols:
 - i. IPv4
 - ii. IPv6
 - iii. ICMP
 - iv. IPsec
 - v. OSPF
 - vi. RIP
 - vii. BGP
6. Data Link Layer: This is the 2nd layer. It provides communication between two devices that are directly connected.
- a. Functionality:
 - i. Converts packets into frames.
 - ii. Uses MAC addresses.
 - iii. Provides node-to-node delivery.
 - iv. Detects errors in transmitted frames.
 - v. Controls access to the transmission medium.
 - vi. Provides framing.
 - b. Data Format: A frame generally contains: Header + Data + Trailer. The header contains information such as source and destination MAC addresses.
 - c. Protocols:
 - i. Ethernet
 - ii. Wi-Fi (IEEE 802.11)
 - iii. PPP
 - iv. HDLC

7. Physical Layer: This is the 1st and lowest layer. It is responsible for physically transmitting data.
- a. Functionality:
 - i. Transmits raw bits.
 - ii. Converts bits into electrical, optical or radio signals.
 - iii. Defines cables and connectors.
 - iv. Defines signal levels, frequencies and data rates.
 - v. Establishes the physical connection between devices.
 - b. Data Format: Bits (0 and 1)
 - c. Protocols:
 - i. Ethernet physical standards
 - ii. USB
 - iii. Bluetooth physical layer
 - iv. DSL
 - v. Fiber-optic standards

Q3. What are the different Error Detection methods? Explain each with the help of proper examples. (3 M)

Ans. Error detection is a technique used in computer networks to determine whether the data received by the receiver is the same as the data sent by the sender. During data transmission, errors may occur due to noise, interference, or other communication problems. Error detection methods add extra bits to the original data so that the receiver can identify whether an error has occurred.

The error detection methods are:

1. Parity Check
2. Checksum
3. Cyclic Redundancy Check (CRC)

Parity Check

Parity check is one of the simplest error detection methods. In this method, an additional bit called a parity bit is added to the data. The parity bit is used to make the total number of 1s either even or odd.

There are two types of parity: even and odd parity.

Even Parity

In even parity, the total number of 1s, including the parity bit, must be even.

For example, consider the data: 1011010.

The number of 1s is 4, which is already even. Therefore, the parity bit is 0.

Data = 1011010

Parity bit = 0

Transmitted data = 10110100

Suppose one bit changes during transmission:

Sent: 10110100

Received: 10111100

The received data contains 5 ones, which is odd. Therefore, the receiver detects that an error has occurred.

Odd Parity

In odd parity, the total number of 1s, including the parity bit, must be odd.

For example: Data = 1100101

There are 4 ones in the data. Therefore, the parity bit must be 1 to make the total number of 1s odd.

Data = 1100101

Parity bit = 1

Transmitted data = 11001011

Advantages:

- It is simple and easy to implement.
- It requires very little additional processing.

Disadvantage: It cannot reliably detect errors when an even number of bits are changed.

Checksum

Checksum is an error detection method in which the data is divided into fixed-size blocks. These blocks are added together using binary addition. The complement of the final sum is called the checksum. The checksum is transmitted along with the original data. At the receiver, the same calculation is performed. If the result indicates an error, the receiver knows that the data has been corrupted.

Example: Suppose the data is divided into three 4-bit blocks:

1001

0110

1010

Add the three blocks. The carry will be added back to the result. The answer is 1010.

Take 1's complement of it: 0101. This is the checksum.

The sender will transmit:

1001 | 0110 | 1010 | 0101

At the receiver, all the blocks including the checksum are added. If the final result is all 1s, the data is considered error-free. Otherwise, an error is detected.

Advantages:

- Simple to implement.
- More effective than a simple parity check.
- Commonly used in networking and communication systems.

Disadvantage: It is less powerful than CRC for detecting certain types of errors

Cyclic Redundancy Check (CRC)

Cyclic Redundancy Check (CRC) is a powerful error detection technique commonly used in computer networks. It uses binary division and XOR operations to detect errors. In CRC, the sender and receiver agree on a value called the generator polynomial.

Steps of CRC:

1. The sender takes the original data.
2. Zeros are appended to the data.

3. The resulting binary number is divided by the generator using XOR division.
4. The remainder obtained from the division is called the CRC.
5. The CRC is appended to the original data and transmitted.
6. The receiver divides the received data by the same generator.
7. If the remainder is zero, the data is considered error-free. If the remainder is non-zero, an error is detected.

Example:

Data = 1001

Generator = 1011

The generator has 4 bits, so 3 zeros are appended to the data:

1001 000

Do XOR division: $1001000 \div 1011$

Remainder obtained: 001

CRC is 001.

The CRC is appended to the original data:

Original data = 1001

CRC = 001

Transmitted data = 1001001

At the receiver, the received data is divided by the same generator:

$1001001 \div 1011$

If the remainder is: 000, then no error is detected. If the remainder is non-zero, it indicates that an error occurred during transmission.

Advantages of CRC:

- It is highly effective in detecting errors.
- It can detect many types of multiple-bit errors.
- It is widely used in computer networks and digital communication systems.

Disadvantage: It is more complex than parity check and checksum.